

Companion document: partial results, conjectures, and computational evidence

Supplement to *The Universal Right-Triangle Reflection Sequence*

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Preface. This companion document collects partial results, conjectures, and computational evidence supporting the program of the main paper `paper.tex`. **Nothing here is claimed as a theorem.** All results are explicitly conditional or empirical: each is labelled either as a *conjecture* (a target statement we believe but have not proved), as a *conditional proposition* (a deduction valid under a hypothesis that is itself only empirically verified at a finite horizon), or as a *numerical observation* (a finite computational record). Cross-references to theorems of the main paper use the same labels.

1. The transcendence program for β_2 : conjecture and three attack routes

Define the asymptotic ratio

$$\beta_2 := \lim_{d \rightarrow \infty} \frac{u_{d+1}}{u_d}$$

of the universal right-triangle reflection sequence. The limit exists and equals

$$\beta_2 = 1.4916177871143742268\dots = 1/\sqrt{q^*},$$

where $q^* = 0.44945363055894\dots$ is the least positive root of $\Sigma_1(q) = 1$; this is unconditional, and the singularity is a simple pole ($\Sigma_0(q^*) \neq 0$). Only the arithmetic nature of β_2 remains open.

Conjecture 1 (Transcendence of β_2). *β_2 is transcendental over $\mathbb{Q}$, and the generating function $U(t) = \sum_{d \geq 0} u_d t^d$ is transcendental over $\mathbb{Q}(t)$.*

We collect three independent attack routes. Each rests on a different finite-horizon hypothesis that is empirically verified and structurally attackable; the routes are logically independent and would each individually settle Conjecture 1.

1.1 Route 1: Christol mod- p via the explicit cyclic-module structure

Let $\rho_{\text{abs}} : W \rightarrow Q \ltimes M$ be the universal realisation of the main paper (§4) with $Q = D_\infty \times \mathbb{Z}/2$ and $M = \mathbb{Z}[Q]/(Y + 1, c + 1)$. For each prime p , write

$$S_p := \{ (\pi(g), m \bmod p) : (g, m) = \rho_{\text{abs}}(w), w \in W \} \subset Q \times (M/pM).$$

Proposition 1 (Mod- p state space is infinite). *For every prime p , $|S_p| = \infty$. In particular the infinite family $\{((XY)^n, t_n) : n \geq 1\}$ (with t_n the translation part of $\rho_{\text{abs}}((R_0 R_2)^n) \bmod p$) consists of pairwise distinct elements of $Q \times (M/pM)$.*

Sketch. XY has infinite order in D_∞ , so the Q -components $(XY)^n$ are pairwise distinct. This alone suffices for infinity of S_p via projection. The stronger separation on the (M/pM) -coordinate, used in Route 1 below, follows from the witness monomial $(XY)^{n-1}X \cdot h$ surviving reduction modulo p by the cyclic-module relations $(Y+1)h = (c+1)h = 0$.

Numerical record. An exact-arithmetic Rust BFS in $\mathbb{F}_p[Q]$ enumerates $S_p(d) := \{\rho_{\text{abs}}(w) \bmod p : |w|_W \leq d\}$ at three primes:

p	$d_{\max}$	$ S_p(d_{\max}) $	β_p (ratio fit, last 10 depths)
3	42	65 428 964	1.4681
5	40	62 736 544	1.5048
7	40	65 724 148	1.5077

The implementation uses the fixed-degree truncation justified by $(Y+1)h = (c+1)h = 0$, which bounds the support of any reachable $m \in M/pM$ by $O(d)$ monomials.

Proposition 2 (U transcendence: Christol route, conditional). *Conditional on the Myhill–Nerode separation hypothesis (the states associated to $\{((XY)^n, t_n)\}_{n \geq 1}$ in the p -DFAO for $u_d \bmod p$ are pairwise distinct future-equivalence classes), $U(t)$ is transcendental over $\mathbb{Q}(t)$.*

Sketch. Under the separation hypothesis the p -DFAO has infinitely many states, hence (Allouche–Shallit Thm. 6.6.2) the p -kernel of $U(t) \bmod p$ is infinite [6], so by Christol [4, 5] $U(t) \bmod p$ is not algebraic over $\mathbb{F}_p(t)$. Adamczewski–Bell 2017 lifts this to non-algebraicity of $U(t)$ over $\mathbb{Q}(t)$. The conclusion is transcendence of U , and it stops there; see Remark 2.

1.2 Route 2: D-finite exclusion

Proposition 3 (No D-finite recurrence at low complexity). *Let (k, m) satisfy $k \leq 9$, $m \leq 7$ and the over-determination condition*

$$(k+1)(m+1) < 43 - k, \quad (1)$$

the right side being the number of equations available from $u_0, \dots, u_{42}$ at order k . There are 52 such pairs, and on none of them does a D-finite (P-recursive) recurrence of order k with polynomial-in- d coefficients of degree m exist (exact integer linear algebra modulo the Mersenne prime $2^{61} - 1$).

Remark 1 (Why the guard is not optional). Condition (1) is what makes a negative result mean anything. Of the 72 pairs with $k \leq 9$ and $m \leq 7$, exactly 20 violate it, and on those the linear system is underdetermined, so a solution exists for *any* input sequence whatever and no exclusion can be read off. The extreme case is $(k, m) = (9, 7)$, with 80 unknowns against 34 equations. An earlier version of this proposition quantified over all 47 tested pairs without the guard and was therefore false as stated; the companion [10] imposes the corresponding condition explicitly. The same remark applies to any extension of the search: raising k or m without also lengthening the sequence buys nothing.

The underlying fact is elementary and is machine-checked in `lean/with_mathlib/OverDetermination.lean`. A homogeneous system with strictly more unknowns than equations has a nontrivial solution for *every* coefficient matrix (`exists_nonzero_solution`), so a search over such a shape reports the existence of a relation whatever the input data are. Contrapositively, an exclusion that is correct forces unknowns $\leq$ equations (`le_of_exclusion`); that is the guard, and its failure convicts the claim. The same file proves the two counts quoted here, `vacuous_card` = 20 of the 72 pairs and `252 > 38` for the nonlinear shape, and records that the guard is necessary but not sufficient (`guard_not_sufficient`): passing it still leaves the rank of the coefficient matrix to be computed.

Proposition 4 (*U* transcendence: D-finite route, conditional). *Conditional on extending Proposition 3 to all $(k, m) \in \mathbb{N}^2$, $U(t)$ is transcendental over $\mathbb{Q}(t)$.*

Remark 2 (The direction of the Flajolet–Sedgewick step, and what these routes do not give). Earlier versions of both routes ended with the sentence that Flajolet–Sedgewick, *Analytic Combinatorics* Thm. VII.7, “implies β_2 transcendental”. That inference is backwards. Theorem VII.7 says that an algebraic generating function has finitely many singularities, all algebraic branch points or poles, so that *algebraic U forces algebraic β_2* . Contrapositively it lets one deduce transcendence of *U* from transcendence of β_2 , which is the opposite of what is wanted here.

The converse is false, so no repair of the wording will do. Take

$$U_*(t) = \frac{1}{1-2t} + \sum_{d \geq 0} t^{2^d} \in \mathbb{Z}[[t]].$$

Its coefficients are 2^n plus the indicator that n is a power of two, so the growth rate is 2, an algebraic number, while the lacunary part gives the unit circle as a natural boundary and U_* is transcendental over $\mathbb{Q}(t)$. Transcendence of the series is thus compatible with an algebraic growth rate.

Consequently neither route above reaches β_2 , and neither does Route 3. What they reach, conditionally, is transcendence of *U*, and that is now known unconditionally by the companion paper [10], which supersedes them. Transcendence of β_2 itself remains open, and this note contains no approach to it.

Sketch. By Stanley *Enumerative Combinatorics* Vol. 2 Thm. 6.4.6, every power series algebraic over $\mathbb{Q}(t)$ is D-finite. Excluding D-finite at all (k, m) therefore excludes algebraic, and gives transcendence of *U*; it says nothing about β_2 , by Remark 2. The natural structural target for the extension is a Gröbner-basis argument on the skew-polynomial ring $\mathbb{Q}\langle n, S \rangle$ acting against the cyclic-module relations $(Y + 1)h = (c + 1)h = 0$.

1.3 Route 3: Mahler structure

Conjecture 2 (*U(t)* is not a class-(1) Mahler function). *U(t) is not a Mahler function of class (1) in the sense of Adamczewski–Faverjon 2020; for the general theory of Mahler functions see [7, 8], and for the transcendence machinery that a positive answer would feed, [9].*

Were Conjecture 2 established, the Adamczewski–Faverjon framework would yield a third independent route to transcendence of β_2 , parallel to Propositions 2 and 4.

1.4 Joint picture

The three conditionals are logically independent, and all three bear on the transcendence of *U* rather than of β_2 . Since [10] now settles transcendence of *U* unconditionally, they are retained only as a record of routes that were explored. The second half of Conjecture 1, transcendence of β_2 , is untouched by any of them.

2. Structural exclusions for $U(t)$ at low complexity

The following finite linear-algebra exclusions on $u_0, \dots, u_{42}$ constrain the form of $U(t)$ at the verifiable horizon.

Proposition 5 (No low-order linear recurrence). *The sequence $u_0, \dots, u_{38}$ satisfies no linear recurrence over $\mathbb{Q}$ of order ≤ 19 . (Equivalently: for every $k \leq 19$, the system $u_n = \sum_{j=1}^k c_j u_{n-j}$ has empty rational solution set on the available terms.)*

Proposition 6 (No nonlinear or modular structure at low complexity). *On $u_0, \dots, u_{42}$:*

- (i) No D -finite recurrence with $k \leq 9$ and $m \leq 7$ (Proposition 3).
- (ii) Withdrawn. An earlier version asserted that no nonlinear polynomial relation $\sum_{\alpha} q_{\alpha}(n) u_n^{\alpha_0} \cdots u_{n-w}^{\alpha_w} = 0$ holds with lookback $w \leq 5$, total u -degree $|\alpha| \leq 3$ and coefficient degree ≤ 2 . That search is vacuous: the monomials of total degree at most 3 in six variables number $\binom{9}{3} = 84$, so with three coefficient degrees there are 252 unknowns against the 38 equations available from $u_0, \dots, u_{42}$. An underdetermined system has solutions for every input, so nothing is excluded. No nonlinear relation of that shape is testable at all on 43 terms.
- (iii) No non-trivial modular linear recurrence at any modulus $m \in \{3, 4, \dots, 101\}$ over the tested range, and at $m = 2$ the order-2 Fibonacci recurrence $u_d \equiv u_{d-1} + u_{d-2} \pmod{2}$ holds for $d \geq 3$ but not at $d = 2$, where $u_0 + u_1 = 4 \equiv 0$ while $u_2 = 5 \equiv 1$.

Proposition 7 (Coefficient-height gap). $\log u_d = c \cdot d + o(d)$ with $c = \log \beta_2 = 0.39986129 \dots$, this being forced by the growth rate established in [10]; ordinary least squares on $d \in [25, 42]$ returns 0.407927, with $R^2 \geq 0.999$, the gap from $\log \beta_2$ being the usual finite-horizon bias. An earlier version quoted $c \approx 0.404$, which matches neither: $e^{0.404} = 1.49780 \dots$ against $\beta_2 = 1.49161 \dots$

The consequence to be drawn is only the weak one. Since u_d grows exponentially it is unbounded, so it is not the value sequence of any k -automatic integer sequence, automatic sequences taking finitely many values; no height-gap theorem is needed for that. In particular the earlier appeal to a height gap to exclude Mahler functions was misplaced: by Adamczewski, Bell and Smertnig [2] the coefficient sequences of Mahler functions realise five distinct growth behaviours, all of them attained, so exponential growth excludes nothing.

Proposition 8 (Mod-2 algebraicity). $u_d \pmod{2}$ is eventually 3-periodic, with pre-period 1: $u_d \equiv 1$ for $d \not\equiv 0 \pmod{3}$ or $d \leq 2$, and $u_d \equiv 0$ otherwise. It is not purely periodic, since $d = 0$ is $\equiv 0 \pmod{3}$ yet $u_0 \equiv 1$. The order-2 Fibonacci recurrence $u_d \equiv u_{d-1} + u_{d-2} \pmod{2}$ holds for $d \geq 3$, and fails at $d = 2$: $u_0 + u_1 = 4 \equiv 0$ while $u_2 = 5 \equiv 1$. Hence $U(t) \in \mathbb{F}_2[[t]]$ is algebraic over $\mathbb{F}_2(t)$ via a 4-state finite automaton.

Caveat. This is a characteristic-2 statement and does not lift to algebraicity over $\mathbb{Q}(t)$.

3. Universality class: scope and known examples

The mechanism that proves universality in the main paper (the universality theorem of [10]) (a cyclic-ideal identification $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$ with explicit central generator h inside the linear quotient Q of a free product $W = A * B$) admits a natural class of free-product reflection systems.

Conjecture 3 (Open question: scope of the cyclic-ideal mechanism). *Characterise the pairs (W, ρ) for which the cyclic-ideal mechanism of the universality theorem of [10] applies and yields universality of the orbit-growth sequence across the family of geometric realisations.*

Two examples are currently known.

Example A: right triangles. $A = V_4$, $B = \mathbb{Z}/2$, $Q = D_{\infty} \times \mathbb{Z}/2$, $h = (c - 1)(Y - 1)$. This is the main paper's the universality theorem of [10].

Example B: hyperbolic two-right-angle quadrilaterals. $A = B = V_4$, $W = V_4 * V_4$, $Q = D_{\infty} \times \mathbb{Z}/2$, $h = c - 1$ (no $(Y - 1)$ factor needed: centrality of c alone suffices when both factors of the free product contain c). This example is recorded as a claim only: we have neither a proof that the mechanism applies nor a computational artifact for it, and it should not be relied on.

Cases where the mechanism is structurally blocked. Acute and obtuse triangle reflection groups have no central involution playing the role of c . Equilateral and other triangle groups with all finite Coxeter labels lack the free-product structure. The rectangle group $D_{\infty} \times D_{\infty}$ is

a direct product, not a free product. Right-corner orthoschemes in dimension $n \geq 3$ carry cross-relations $(R_i R_j)^2 = 1$ between generators of distinct free-product factors, again destroying the $W = A * B$ structure; universality in this cyclic-ideal sense is therefore specific to dimension 2.

4. Structural reading of universality (algebraic-channel BFS)

Fix $\rho_{a,b} : W \rightarrow \text{Aff}(\mathbb{R}^2)$. Let Y denote the linear part of $\rho_{a,b}(R_2)$ and v_2 its translation part. The identity $Yv_2 = -v_2$ holds for every (a, b) and is the image of a single canonical word $w_\star \in W$. Let N_{alg} be the normal closure of $w_\star$ in W .

Numerical observation 1 (Empirical algebraic form of universality). Through depth 22, on the four test right triangles $(3, 4)$, $(5, 12)$, $(1, 2)$, $(7, 24)$, and within the horizon in which the companion's universality statement itself holds (it is proved to depth 32 and fails at depth 33 for $(1, 2)$), the equivalence relation on canonical Coxeter words generated by $Yv_2 + v_2$ in $\mathbb{Z}[D_\infty]$ exactly recovers the affine equivalence relation. Equivalently: $N(T) = N_{\text{alg}}$ on the depth- ≤ 22 filtration for every tested T .

A parallel BFS verifies this with cumulative state count 118 938 at depth 22 in every channel: four channels for the four test triangles above and a fifth for the algebraic model N_{alg} against which they are compared. Reading: the eight length-10 affine relations of the symbolic-verification theorem of [10] are not sporadic length-10 coincidences but the first eight consequences of the single quadratic relation $Yv_2 + v_2 = 0$, made visible by the canonical-word filtration.

5. Empirical evidence against finite confluent rewriting

Knuth–Bendix experiment. GAP KBMAG ran Knuth–Bendix completion on W together with the eight length-10 affine relations of the symbolic-verification theorem of [10]; the procedure produced 32 730 derived rules in approximately 8 seconds of `kbprog` computation before hitting the equation-count cap, never reaching confluence. This is empirical evidence (not a proof) that W/N has no finite confluent rewriting system in shortlex on the standard generators.

Antolín–Ciobanu route (structurally blocked). Antolín and Ciobanu prove that the conjugacy-growth series of a non-elementary acylindrically hyperbolic group is *transcendental* [1]; an earlier version of this paragraph called it a rationality theorem, which inverts the result, and attributed it to the wrong journal. Either way it does not apply here, and for a reason that is about the group and not about the series: the image of $W/\ker(\rho_{\text{abs}})$ lies inside $Q \rtimes M \subseteq (\mathbb{Z} \wr \mathbb{Z}) \rtimes V_4$, a virtually metabelian group, which is not acylindrically hyperbolic. Note also that $\mathbb{Z} \wr \mathbb{Z}$ is *not* finitely presented, by Baumslag; the earlier description of the ambient group as finitely presented was wrong, and is not needed for the conclusion.

6. Class A and Class B in 3D

The 3D right-corner orthoscheme reflection groups fall into three faithfulness regimes by leg-symmetry. The Class C case, pairwise distinct legs, is treated in the n -dimensional companion [3], where the exclusion of rank-two relations is proved and faithfulness itself is stated as a conjecture; it is not a theorem, and nothing below assumes it.

Class A ($a = b = c$, S_3 -symmetric, polynomial growth). The $(1, 1, 1)$ cube-corner orthoscheme is the fundamental chamber of the affine Weyl group $\tilde{C}_3$. For $n \geq 1$, $u_n = (8n^2 + c_{n \bmod 5})/5$ with $c = (10, 12, 13, 13, 12)$, with asymptotic $u_n \sim (8/5)n^2$. The closed form, its integrality and its agreement with the Bott series of $\tilde{C}_3$ hold for $n \geq 1$; in particular $u_1 = 4$ and $u_2 = 9$. Order-7 linear recurrence with characteristic polynomial $(x - 1)^3 \Phi_5(x)$, obtained

from the series $W_A(t) = N(t)/((1-t)^3\Phi_5(t))$ with $N(t) = 2 + 3t^2 + 3t^3 + 3t^4 + t^5 + 4t^6$. Two cautions about that display. Its constant term is $N(0) = 2$, whereas an orbit-growth series has constant term 1, so as written it is not the growth series but that series plus one; the closed form for $n \geq 1$ is unaffected, and the discrepancy sits entirely in degree 0. And “parabolic Poincaré series” is a misnomer for it. This is a clean restatement of standard affine-Weyl-group material (Bourbaki Ch. VI §4; Humphreys §4.7) and is recorded here for completeness.

Class B (exactly two equal legs, intermediate exponential rate). Through depth 18 the sequence on $(1, 1, 2)$ is 1, 4, 9, 18, 35, 67, 129, 249, 480, 925, 1783, 3437, 6625, 12770, 24591, 47298, 90948, 174939, 33 with empirical ratio settling around ≈ 1.92 . No order- ≤ 9 linear recurrence over $\mathbb{Q}$ fits the depth-18 data. No closed form is known.

7. Other deferred material

The following items were deferred from the main paper to this supplement and remain available in earlier drafts of the project:

- Numerical and structural analysis of the representation gap β_n vs. r_n across dimensions.
- Collision-class enumeration to depth 30, the structural sequence battery, the $\sqrt{3}$ -asymptotic of $\delta_d - 2z_d$, and the crossover $\delta_d > u_d$ for $d \geq 24$.
- $\{3, 4, 5\}$ -cascade observation for maximal outerplanar graphs on $n \leq 15$ vertices, the slack/denominator/precision analysis, and the cascade-existence conjecture.
- The standard 3-simplex remark and merged length-10-relation cascades from Pythagorean triples sharing a common edge length.
- Per-script hardware envelopes and runtime tables for the BFS implementations.

These items are unchanged from earlier supplementary drafts and are recorded here as a placeholder; consult the project repository for the detailed records.

References to literature are as in the main paper. The additional references invoked by the conditional routes above are:

- G. Christol, *Ensembles presque périodiques k-reconnaissables*, TCS 9 (1979), 141–145.
- J.-P. Allouche, J. Shallit, *Automatic Sequences*, Cambridge University Press, 2003.
- B. Adamczewski, J. P. Bell, *On the algebraic dependence of p-automatic series and G-functions over function fields of positive characteristic*, J. Number Theory 180 (2017), 559–586.
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- R. P. Stanley, *Enumerative Combinatorics*, Vol. 2, Cambridge University Press, 1999. Thm. 6.4.6 (algebraic $\Rightarrow$ D-finite).

- P. Flajolet, R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Thm. VII.7.
- Y. Antolín, L. Ciobanu, *Formal conjugacy growth series and language complexity*, Math. Z. 285 (2017), no. 3, 1015–1041.

8. Machine verification

The finite and combinatorial content of the propositions above is machine-checked in Lean 4. Two files in the Mathlib-free project carry it, so they compile in core Lean with no dependency fetch; a clean `lake build` rechecks the whole project, including them, in about four minutes.

`PaperExtraMod2.lean` discharges Proposition 8 on the 39 published terms: the closed form $u_d \bmod 2$, eventual 3-periodicity with pre-period 1, the failure of pure periodicity at $d = 0$, the order-2 Fibonacci recurrence for $d \geq 3$, and its failure at $d = 2$. It also records the agreement with Fibonacci for $1 \leq n \leq 9$ and the first deviation $u_{10} = 225$. These same two recurrence facts are the mod-2 half of Proposition 6(iii).

`PaperExtraCounts.lean` verifies the search grid of Proposition 3: that the over-determination condition (1) selects exactly 52 pairs, and that every selected pair has strictly fewer unknowns than equations. It also verifies the arithmetic behind the withdrawal in Proposition 6(ii), namely $\binom{9}{3} = 84$ monomials, 252 unknowns and 38 equations, and it proves the general form of the argument used in Proposition 7: for any notion of automaticity whose sequences are finitely valued, an unbounded sequence is not automatic. The finitely-valued property appears as an explicit hypothesis rather than being assumed silently.

`NoRecurrence.lean` certifies Proposition 5 outright. For each order k with $1 \leq k \leq 19$ the system of $39 - k$ equations in k unknowns is shown to have no rational solution by exhibiting a Farkas witness: an integer vector w with $w^\top A = 0$ and $w^\top b \neq 0$, which forbids a solution since any rational c would give $w^\top b = w^\top (Ac) = 0$. The witnesses were computed once in exact GMP arithmetic by the Rust tool `code/zeta_probe/tools/norec`; Lean re-runs none of that linear algebra and verifies only the two defining integer dot products, so a reader need not trust the search that produced them. All nineteen witnesses are small, at most 22 digits per entry, so the check is kernel evaluation and carries no compiler-evaluation axiom. The Farkas step itself is proved for arbitrary integer data, so it is visibly independent of this sequence.

One convention is worth recording, since the statement of Proposition 3 leaves it implicit. The count 52 is for $k \geq 1$, that is for recurrences of order at least one. Admitting $k = 0$ gives 60 pairs and requiring $m \geq 1$ gives 50; only $1 \leq k \leq 9$, $0 \leq m \leq 7$ reproduces the published figure. The Lean file verifies both counts, so the convention is fixed by the formalization rather than by the prose.

What is not formalized is stated plainly in each file. The D-finite exclusion computations of Proposition 3, being exact linear algebra over 43 terms at 52 parameter pairs, are not re-run in Lean, and neither is the passage from eventual periodicity to algebraicity over $\mathbb{F}_2(t)$, which needs formal power series; nor the unboundedness of u_d , which is a statement about all d and rests on the growth rate of [10]. Strict increase across the published range is verified, and is evidence for unboundedness rather than a proof of it. The regression numbers in Proposition 7 are numerics and are not formalized.

References

- [1] Y. Antolín and L. Ciobanu, *Formal conjugacy growth in acylindrically hyperbolic groups*, Int. Math. Res. Not. IMRN **2017**, no. 1, 121–157.

- [2] B. Adamczewski, J. P. Bell and D. Smertnig, *A height gap theorem for coefficients of Mahler functions*, J. Eur. Math. Soc. **23** (2021), 2379–2398.
- [3] V. Bonfioli, *A dimension-independent collision-depth invariant for right-corner orthoscheme reflection groups*, companion paper.
- [4] G. Christol, *Ensembles presque périodiques k -reconnaissables*, Theoret. Comput. Sci. **9** (1979), 141–145.
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- [9] Yu. V. Nesterenko, *Modular functions and transcendence questions*, Sb. Math. **187** (1996), 1319–1348.
- [10] V. Bonfioli, *Transcendence of the right-triangle reflection growth series*, companion paper; code and formalization at github.com/ElVec1o/pythagorean-reflection-sequence.